# Planes of Symmetry

Exam code: 0607

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**Written by:** Amber
**Reviewed by:** Dan Finlay
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## Planes of symmetry

### What is a plane of symmetry?

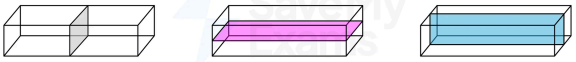
- A **plane** is a flat surface that can be any 2D shape
- A **plane of symmetry** is a **plane** that splits a 3D shape into two **congruent** (identical) halves
- If a 3D shape has a plane of symmetry, it has **reflection symmetry**
  - The two congruent halves are identical, mirror images of each other

- All **prisms** have at least one plane of symmetry
  - **Cubes** have 9 planes of symmetry
  - **Cuboids** have 3 planes of symmetry
  - **Cylinders** have an infinite number of planes of symmetry
  - The number of planes of symmetry in other prisms will be equal to the number of lines of symmetry in its cross-section plus 1

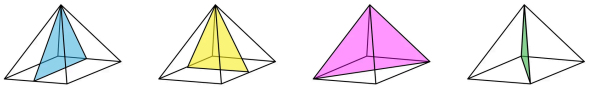
- **Pyramids** can have planes of symmetry too
  - The number of planes of symmetry in pyramids will be equal to the number of lines of symmetry in its 2D base
  - If the base of the pyramid is a **regular polygon** of *n* sides, it will have *n* planes of symmetry

A cube has 9 planes of symmetry

A cuboid has 3 planes of symmetry



A square based pyramid has 4 planes of symmetry



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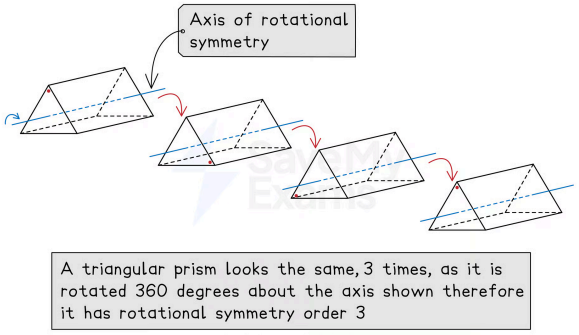
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- 3D shapes are able to be **rotated around different axes**
  - Depending on which **axis** the shape is rotated around, **3D shapes can have rotational symmetry**
- Recall that **rotational symmetry** is how many times the shape looks the same (congruent) when rotated through 360 degrees
  - See the example of the triangular prism where the cross-section is an **equilateral triangle**

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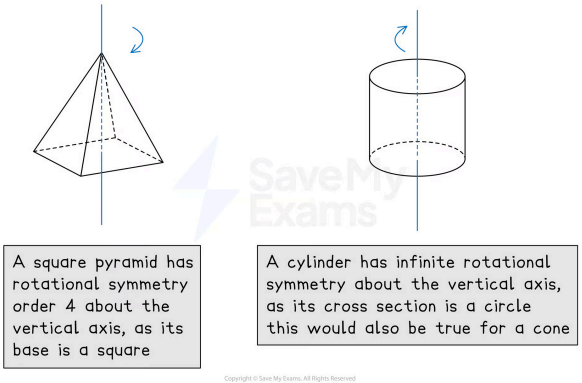


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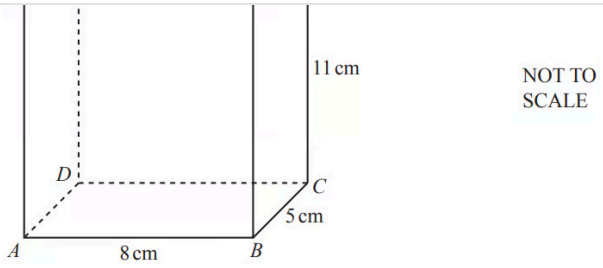
## Examiner Tips and Tricks

- If you're unsure in the exam, consider the properties of the 3D shape.
  - Is it a **prism** or a **pyramid**?
  - How many **lines of symmetry** are there in the **2D faces** or **cross-section**?

## Worked Example

The diagram below shows a cuboid of length 8 cm, width 5 cm and height 11 cm.

Write down the number of planes of symmetry of this cuboid.




**Answer:**

A plane of symmetry is where a shape can be "sliced" such that it is symmetrical

A cuboid with three different pairs of opposite rectangles has 3 planes of symmetry

**3 planes of symmetry**

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